



MGM's College of Engineering, Nanded

Department of Computer Science & Engineering

“WellNest – Smart Health & Fitness Companion ”

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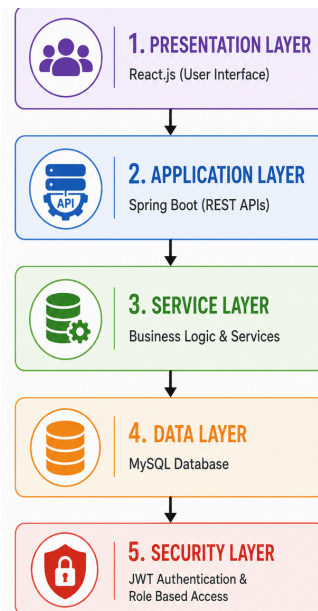
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Introduction:

This project presents WellNest – Smart Health & Fitness Companion, an intelligent digital wellness management system designed to help users monitor and improve their overall health. The system provides features such as hydration tracking, yoga management, walking activity monitoring, BMI calculation, trainer consultation, health blogs, and AI chatbot assistance. It promotes healthy lifestyle habits through personalized tracking, real-time monitoring, and expert guidance. The platform aims to improve user engagement in wellness activities while providing a secure and user-friendly experience.

System Architecture:

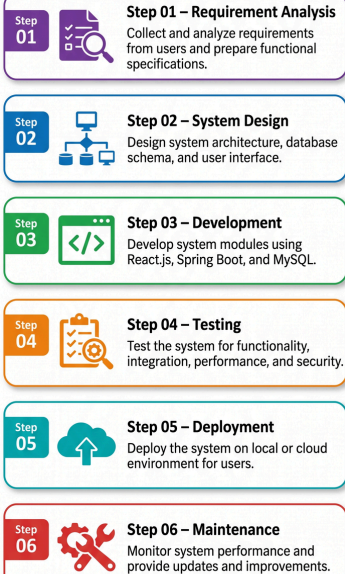
The proposed system follows a layered architecture to ensure scalability, security, and efficient data management. The presentation layer is developed using React.js for user interaction and dashboard management. The application layer uses Spring Boot to process business logic and API requests. MySQL is used as the database layer for storing user records and wellness data. JWT authentication provides secure access control, while integrated services such as trainers, blogs, and AI chatbot support enhance overall functionality.



Methodology:

The proposed system follows a structured development methodology for building an efficient digital wellness management platform. The process begins with requirement analysis and system design, architecture are identified. The system is then developed using React.js, Spring Boot, and MySQL to implement all wellness management modules. After development, testing is performed to verify functionality, security, and performance. Finally, the application is deployed and maintained to ensure reliable operation and continuous improvement of the platform.

METHODOLOGY – WELLNEST SYSTEM



Conclusion:

WellNest – Smart Health & Fitness Companion provides a digital wellness solution for monitoring and improving personal health. The platform integrates hydration tracking, yoga management, walking activity monitoring, BMI calculation, trainer consultation, wellness blogs, and AI chatbot support within a single system. The application ensures secure data management and an interactive user experience.

Maps to: -
PO-1, PO-2,
PO-4, PO-5,
PO-7, PO-8,
PO-10, PO-11,
PSO-1, PSO-2,
PSO-3

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